



$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \varphi)$$

$$a = \frac{d^2x}{dt^2} = -\omega^2 A \cos(\omega t + \varphi) = -\omega^2 x$$

结论：简谐振动的动力学特征

$$1) F = -kx$$

$$2) a = -\omega^2 x \quad \text{其中 } \omega \text{ 为常数}$$

$$3) \frac{d^2x}{dt^2} + \omega^2 x = 0$$

它的解即为 $x = A \cos(\omega t + \varphi)$

以上三种特征, 任意满足一种, 则可说该物体做简谐振动

$$\begin{aligned} F &= -kx \\ a &= -\omega^2 x \\ W &= \sqrt{\frac{k}{m}} \\ T &= 2\pi \sqrt{\frac{m}{k}} \\ A &= \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}} \\ \tan \varphi &= -\frac{v_0}{\omega x_0} \end{aligned}$$

► 周期和频率：由振动系统的固有性质决定

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$$

$$\tan \varphi = -\frac{v_0}{\omega x_0}$$

$$T = \frac{2\pi}{\omega} \quad \nu = \frac{1}{T}$$

$$v = \omega R$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

► 振幅和初相：由初始条件决定

$$\begin{cases} A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}} \\ \tan \varphi = -\frac{v_0}{\omega x_0} \end{cases}$$

$$\begin{aligned} \begin{cases} x_0 = A \cos \varphi \\ v_0 = A \omega \sin \varphi \end{cases} & \Leftrightarrow \begin{cases} \sin \varphi = -\frac{v_0}{A \omega} \\ \cos \varphi = \frac{x_0}{A} \end{cases} \end{aligned}$$

$$W = \sqrt{\frac{k}{m}}$$

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$$

$$\tan \varphi = -\frac{v_0}{\omega x_0}$$

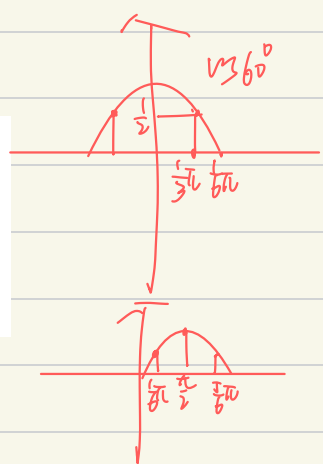
单摆 $\omega = \sqrt{\frac{g}{L}}$

$\sqrt{\frac{12}{m}}$

$$E_p = \frac{1}{2} kx^2 = \frac{1}{2} kA^2 \cos^2(\omega t + \varphi)$$

$$E_k = \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \varphi)$$

$$E_k = \frac{1}{2} kA^2 \sin^2(\omega t + \varphi)$$



$E = E_k + E_p = \frac{1}{2} kA^2$

计算表明,在振动过程中弹簧振子的动能和弹性势能相互转化,如图 4-9(a) 所示. 在平衡位置 $x=0$ 处,势能为零, $x = \pm A$ 处,势能最大,动能为零,但是其总能量不随时间变化,势能相互转化,但总机械能守恒.

图 4-9(b) 所示的抛物线是弹簧振子的机械能曲线. 横轴为位移 x , 纵轴为机械能 E .

$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$
 $\tan \phi = - \frac{v_0}{\omega x_0}$

振动: 同频率同方向:

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_1 - \varphi_2)}$$

$$\tan\varphi = \frac{A_1\sin\varphi_1 + A_2\sin\varphi_2}{A_1\cos\varphi_1 + A_2\cos\varphi_2}$$

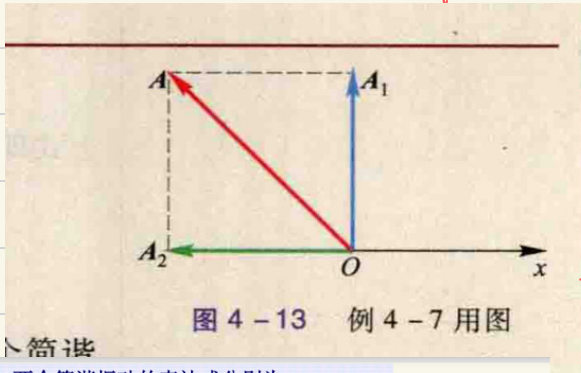


图 4-13 例 4-7 用图

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_1 - \varphi_2)}$$

$$\tan\varphi = \frac{\sin\varphi_1 A_1 + \sin\varphi_2 A_2}{\cos\varphi_1 A_1 + \cos\varphi_2 A_2}$$

例4-3 已知两个简谐振动的表达式分别为

$$x_1 = 2\cos(10\pi t + \frac{\pi}{2})$$

$$x_2 = 2\cos(10\pi t - \pi)$$

(1) 求合振动的表达式;

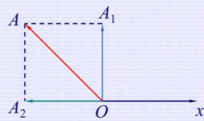
(2) 若 $x_3 = 3\cos(10\pi t + \theta)$, 则 θ 为何值时, 三振动叠加后, 合振动的振幅最大? 则 θ 为何值时, 三振动叠加后, 合振动的振幅最小?

解:

$$(1) A = \sqrt{2}A_1 = 2\sqrt{2}\text{m}$$

$$\varphi = \frac{3\pi}{4}$$

$$x = 2\sqrt{2}\cos(10\pi t + \frac{3\pi}{4})$$



$$(2) \theta = 2k\pi + \frac{3\pi}{4} \text{ 时, 合振动振幅最大, } A_{\max} = (3 + 2\sqrt{2})\text{m}$$

$$\theta = 2k\pi + \frac{7\pi}{4} \text{ 时, 合振动振幅最小, } A_{\min} = (3 - 2\sqrt{2})\text{m}$$

$$A = \sqrt{(A_2)^2 + 3^2 + 2 \times 3 \times A_2 \cos(\theta - \frac{3\pi}{4})}$$

$$\varphi_2 - \varphi_1 = \begin{cases} 2k\pi & \text{最大} \\ 0 & \text{最小} \end{cases}$$

= +1 时 A 最大
= -1 时 A 最小

1.2 横波与纵波

横波: 质元的振动方向与波的传播方向垂直

纵波: 质元的振动方向与波的传播方向平行

➤ 横波只能在固体中传播，而纵波可以在固体、液体和气体中传播

机械波 (两个要求):
{ 波源
弹性介质

波速 $\rightarrow$ 介质
频率 $\rightarrow$ 波源

$$v = \frac{1}{T}$$

$$u = \frac{\lambda}{T} = \lambda v$$

$$\left\{ \begin{array}{l} v = \frac{1}{T} \\ u = \frac{\lambda}{T} \end{array} \right.$$

$$y = A \cos \left[\omega \left(t - \frac{x}{u} \right) + \varphi \right] \quad \begin{array}{l} - \text{为} \rightarrow \\ + \text{为} \leftarrow \end{array}$$

$$y = A \cos \left[\omega t - \frac{2\pi x}{\lambda} + \varphi \right]$$

$$y = A \cos \left[2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) + \varphi \right]$$

动能 = 势能 动 + 势 = 机

$$y = A \cos \left(\omega \left(t - \frac{x}{u} \right) + \varphi \right) \quad y = A \cos \left(\frac{2\pi}{T} \left(t - \frac{x}{u} \right) + \varphi \right)$$

$$y = A \cos \left(\omega t - \frac{2\pi x}{\lambda} + \varphi \right) \quad = A \cos \left(2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) + \varphi \right)$$

① 公式 2 页

$$y = A \cos \left(\omega \left(t - \frac{x}{u} \right) + \varphi \right)$$

$$y = A \cos \left(\omega t - 2\pi \frac{x}{\lambda} + \varphi \right)$$

$$y = A \cos \left(\omega \left(t - \frac{x}{u} \right) + \varphi \right)$$

$$a = -\omega^2 x$$

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$$

$$\tan \varphi_0 = -\frac{v_0}{\omega x_0}$$

平均能流 $W = \frac{1}{2} P A \omega^2$

平均能流 $P = W u$

能流密度 $I = \frac{P}{S}$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_1 - \varphi_2)}$$

$$\tan \varphi = \frac{\sin \varphi_1 A_1 + \sin \varphi_2 A_2}{\cos \varphi_1 A_1 + \cos \varphi_2 A_2}$$

$$W = \frac{1}{2} P A \omega^2$$

$$P = W u = \frac{1}{2} P A \omega^2 u$$

$$I = W u = \frac{1}{2} P A \omega^2 u$$

平均能量密度: $\bar{w} = \frac{1}{2} \rho A \omega^2$

$\downarrow \quad \downarrow \quad \downarrow$
 密度 振幅 角速度

平均能流 $\bar{P} = \bar{w} u s = \frac{1}{2} \rho A \omega^2 u s$

$\uparrow \uparrow \uparrow$
 平均能密 线密度 截面积

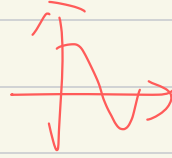
能流密度/波的强度 $I = \frac{\bar{P}}{s} = \frac{1}{2} \rho A \omega^2 u$

干涉: $\Delta \phi = (\phi_2 - \phi_1) - \frac{2\pi}{\lambda} (r_2 - r_1)$

$= \begin{cases} 2k\pi & \text{长} \\ (2k+1)\pi & \text{消} \end{cases}$

$\delta = r_2 - r_1 = \begin{cases} k\lambda & \text{长} \\ (k+\frac{1}{2})\lambda & \text{消} \end{cases}$

$(\phi_2 - \phi_1) - \frac{2\pi}{\lambda} (r_2 - r_1) = \begin{cases} 2k\pi & \text{长} \\ (2k+1)\pi & \text{消} \end{cases}$



反射波的波形刚好重合的时刻作为计时零点,取某个合位移最大的点作为坐标原点 O ,则这两列波可表示为

$$y_1 = A \cos \left(\omega t - \frac{2\pi x}{\lambda} \right)$$

$$y_2 = A \cos \left(\omega t + \frac{2\pi x}{\lambda} \right)$$

↑ 传播
↑ 传播
初相位为0

合成波的表达式为

$$y = y_1 + y_2 = 2A \cos \frac{2\pi x}{\lambda} \cos \omega t \quad (4-78)$$

上式常被称为驻波方程,它就是图 4-54 中合成波(蓝色线)的波动表达式.

$$2A \cos \omega t \cos \frac{2\pi x}{\lambda}$$

$$y = 2A \cos \frac{2\pi x}{\lambda} \cos \omega t$$

> x 轴上各点作简谐振动。
 > 各点振幅随 x 而变化：
 $A = \left| 2A_0 \cos \frac{2\pi x}{\lambda} \right|$

$x_k = (2k+1)\frac{\lambda}{4}, A=0$ (波节)
 $x_k = 2k\frac{\lambda}{4}, A=2A_0$ (波腹)
 $\Delta x = x_{k+1} - x_k = \frac{\lambda}{2}$

> 若相邻波节之间为一段,则同一段中各点的振动位相相同,而相邻段振动的位相相反